GURDHARAM JEET SINGH

Full-Stack Software Engineer • Web Systems & Spatial Frontend Architect

Bathinda / Muktsar, Punjab, India • +91 90411 72159 • gurination1@gmail.com • github.com/gurination1 • linkedin.com/in/gurdharam • gurdharam.com

TECHNICAL SKILLS

Languages & Core: TypeScript, JavaScript (ES6+), Python, Go, SQL, HTML5, CSS3

Frontend & Spatial Web: React 19, Next.js 15 (App Router, Server Components), Tailwind CSS 4, Vite 6, Framer Motion, Lenis Smooth Scroll, Three.js, WebGL2 (Camera Kit), Core Web Vitals (LCP, CLS < 0.02)

Backend & Distributed Systems: Node.js, Express, FastAPI, PostgreSQL, Supabase, MariaDB/SQLite Failover, Redis & BullMQ, WebSockets, REST APIs, Docker, Linux/Bash

Cloud & Platform Infrastructure: Official Meta WhatsApp Cloud API (WABA), GitHub Actions 24/7 CI/CD Fleets, Railway, AWS (S3, Lambda), Exotel Telephony, Gemini AI

KEY PRODUCTION PLATFORMS & SYSTEMS

SOLUM Minerals — Procedural 3D WebGL Bio-Carbon Platform **2026 (Oct) – Present**
Lead Frontend Architect • Next.js 15, Three.js, WebGL2, Tailwind CSS Open Source

- **Procedural 3D Tablet & River Kinematics:** Engineered code-generated Three.js WebGL tablet model with procedural 1024x1024 noise texture synthesis in code, squircle compression grooves, and Webflow-style smooth scrolling.
- **Performance Engineering & Core Web Vitals:** Architected responsive commercial web platform using Next.js 15 App Router, achieving a 99+ Lighthouse performance rating with zero layout shift (CLS < 0.02) and sub-second FCP.

EduCore — Enterprise College Management ERP **2025 – Present**
Lead Full-Stack Architect • React 19, TypeScript, Node.js, Express, MariaDB, Railway Open Source

- **Full-Stack Architecture & Two-Tier Failover:** Engineered enterprise ERP handling student admissions, fee ledgers, and scholarships with automated zero-config DB failover (MariaDB production → local SQLite store).
- **RBAC & Dynamic Form Schema Builder:** Built strict JWT role-based access control across 3 permission tiers, Helmet security headers, Zod validation, and a runtime dynamic form builder for campus workflows.

Dream Heights & Branders — Spatial Architectural & Design Suite **2025 – Present**
Spatial & Frontend Architect • React 18, TypeScript, Three.js, Framer Motion, Lenis Open Source

- **Spatial Architectural Pins & Navigation:** Built interactive luxury residence portal featuring spatial pin exploration, dynamic residence floor-plan modals, and custom interactive architectural emblems.
- **60 FPS Physics & Smooth Momentum Scrolling:** Implemented Lenis inertial scrolling combined with hardware-accelerated Framer Motion transitions and Three.js physics navigation, maintaining silky 60 FPS across viewports.

Enterprise WhatsApp AI & Lead Management CRM **2025 – Present**
Lead Architect • Meta Cloud API, Node.js, Express, Supabase (PostgreSQL) *Production Platform*

- **Official Meta Cloud API Architecture:** Engineered multi-tenant CRM gateway integrating Meta Graph Cloud API (WABA ID: 106690...), handling bi-directional messaging, template dispatch, and HMAC-verified webhooks.
- **24-Hour Autonomous Follow-Up Radar:** Built real-time engagement engine with dynamic cadences (24h/36h/48h/72h), tracking lead conversation states and triggering automated follow-ups via Supabase.

WORK EXPERIENCE

Gurdharam Software Consultancy • Systems & Web Engineer **2025 – Present**
Independent Software Consultancy • Remote *Punjab, India*

- **Telephonic Voice AI Architecture:** Built real-time telephony agent over Exotel WebSockets, SIP, and VoIP, achieving <700ms turn latency across Sarvam AI Indic LLMs, Groq LPU, and Silero VAD barge-in.
- **High-Throughput Webhook Processing:** Architected distributed queue worker system using Redis and BullMQ, reliably processing 25,000+ daily webhook events with automated retry logic and sub-5-minute SLA.

EDUCATION & CREDENTIALS

Baba Farid Group of Institutions (BFGI) — B.Sc. in Computer Science **Expected May 2027**
Core: Data Structures & Algorithms, Database Systems, Computer Networks, Operating Systems *Punjab, India*

Credentials & Verification: Meta Verified Business Entity • Official WhatsApp Business Platform Developer • Sarvam AI Startup Partner • Govt. MSME Registered Enterprise (UDYAM-PB-03-0079744).